

# **SOCIOL 723: Social Statistics II**

Week 1 (Thursday), OLS in Matrix Form

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# Today

The Matrix Form and Its Geometry

Partialling Out in General

Tuesday built everything in scalar form. Today: the same objects again in matrix form, for general understanding, followed by lab.

★ = essential, you must understand this

★ = for understanding, not examined

# Matrix Form

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# Why Bother

- With  $k$  regressors, scalar algebra needs  $k$  normal equations and becomes unreadable. Matrix algebra stays two lines long for any  $k$ .
- The **geometry**, projection onto a subspace, is invisible in scalar form and obvious in matrix form.
- In the literature, most estimators you will meet (fixed effects, IV, 2SLS, ridge) are *written* as small variations on  $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ . We will keep working in scalars, but recognizing this form is what lets you read modern papers.

## But

Nothing in this section is new content. It is the *same* normal equations you derived on Tuesday, written more compactly. The lab that follows practices every piece of linear algebra we use.

## Vectors and Matrices, From Zero ★

No linear algebra is assumed in this course. Here is the part we need, from nothing.

- A **vector** is a list of numbers written in a column.  $X_i$  is the list describing unit  $i$ : one entry per variable, with a 1 on top for the intercept.
- A **matrix** is a rectangular table of numbers: an  $n \times k$  matrix has  $n$  rows and  $k$  columns.
- The **transpose**  $'$  tips a column onto its side: if  $X_i$  is a  $k \times 1$  column, then  $X_i'$  is the same list lying down,  $1 \times k$ .

The one formula to internalize

$$X_i' \beta = \beta_0 \cdot 1 + \beta_1 X_{i1} + \cdots + \beta_{k-1} X_{i,k-1} :$$

a row times a column is a *single number*: multiply matching entries, then add. So  $X_i' \beta$  is nothing but unit  $i$ 's fitted value, written compactly. Whenever you see  $X_i' \beta$ , read it as “the regression line evaluated at unit  $i$ .”

## Multiplying and Inverting

The only two operations we use. **Multiplication** extends the row-times-column rule: entry  $(r, c)$  of  $\mathbf{AB}$  is (row  $r$  of  $\mathbf{A}$ ) times (column  $c$  of  $\mathbf{B}$ ):

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Dimensions must match,  $(n \times k)(k \times 1) = (n \times 1)$ , and order matters:  $\mathbf{AB} \neq \mathbf{BA}$  in general.

**Inversion** is the matrix version of dividing. For a number,  $a^{-1}a = 1$ ; for a square matrix,  $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ , the *identity* matrix (ones on the diagonal, zeros elsewhere), which leaves whatever it multiplies unchanged.

- Just as  $1/a$  fails to exist when  $a = 0$ ,  $\mathbf{A}^{-1}$  fails to exist when the matrix collapses information; that is the “rank” condition, explained shortly.
- You will never invert a matrix by hand here; R’s `solve()` does it. What matters is knowing what the symbol says.

## Stacking the Data

For  $n$  units and  $k$  regressors (including the constant):

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{bmatrix} X'_1 \\ \vdots \\ X'_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1,k-1} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{n,k-1} \end{bmatrix}_{n \times k}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The model, stacked over units:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad \Longleftrightarrow \quad Y_i = X'_i\boldsymbol{\beta} + \epsilon_i, \quad i = 1, \dots, n.$$

Two objects do all the work:  $\mathbf{X}'\mathbf{X} = \sum_i X_i X'_i$  (a  $k \times k$  matrix) and  $\mathbf{X}'\mathbf{y} = \sum_i X_i Y_i$  (a  $k \times 1$  vector).

## Least Squares in Matrix Form: the Expansion

The sum of squared errors is the error vector times itself:

$$S(b) = \sum_i (Y_i - X_i' b)^2 = (\mathbf{y} - \mathbf{X}b)'(\mathbf{y} - \mathbf{X}b).$$

Two transpose rules, used constantly

$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$ , and  $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ : transposing a product *reverses the order*.  
Hence  $(\mathbf{X}b)' = b'\mathbf{X}'$ .

Multiply out, term by term:

$$S(b) = \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}b - b'\mathbf{X}'\mathbf{y} + b'\mathbf{X}'\mathbf{X}b = \mathbf{y}'\mathbf{y} - 2b'\mathbf{X}'\mathbf{y} + b'\mathbf{X}'\mathbf{X}b.$$

- Why do the middle terms merge? Each is  $1 \times 1$ , a single number, and a number equals its own transpose:  $(b'\mathbf{X}'\mathbf{y})' = \mathbf{y}'\mathbf{X}b$ . The two terms are the *same* number written twice.

# Minimizing: Two Matrix Derivatives, From Zero

$\partial S / \partial b$  just stacks the  $k$  partials into a column; two stock rules cover everything:

The rules, with their scalar shadows

$$\frac{\partial(a'b)}{\partial b} = a \quad \text{since } a'b = \sum_j a_j b_j \text{ gives } \partial / \partial b_j = a_j \quad (\text{cf. } \frac{d}{db} ab = a)$$

$$\frac{\partial(b'Ab)}{\partial b} = 2Ab \quad \text{for symmetric } A \quad (\text{cf. } \frac{d}{db} ab^2 = 2ab)$$

Apply them with  $a = \mathbf{X}'\mathbf{y}$  and  $A = \mathbf{X}'\mathbf{X}$  (symmetric):

$$\frac{\partial S(b)}{\partial b} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}b \stackrel{!}{=} 0 \quad \implies \quad \underbrace{\mathbf{X}'\mathbf{X} \hat{\beta}}_{\text{normal equations}} = \mathbf{X}'\mathbf{y}.$$

The *same*  $k$  equations you wrote out one at a time on Tuesday. Why a minimum?

$v'\mathbf{X}'\mathbf{X}v = \|\mathbf{X}v\|^2 \geq 0$  for all  $v$ :  $S$  is a bowl. (Shorthand:  $\mathbf{X}'\mathbf{X} \succeq 0$ , the matrix  $\succeq$ .)

## The Normal Equations, Slowly: the Bivariate Case

Take  $k = 2$  with  $X_i = (1, X_i)'$ , so  $b = (b_0, b_1)'$ . The two building blocks are

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Write out  $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$  one row at a time:

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_i X_i = \sum_i Y_i \iff \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\hat{\beta}_0 \sum_i X_i + \hat{\beta}_1 \sum_i X_i^2 = \sum_i X_i Y_i \iff \sum_i X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

- Row by row, the matrix equation *is* Tuesday's (N1) and (N2): residuals sum to zero, and residuals are orthogonal to the regressor.
- One matrix line,  $k$  scalar equations. That is the entire content of the compact notation: nothing new, everything stacked.

# The Estimator, Existence, and Rank

## What “rank” means, in words

The **rank** of  $\mathbf{X}$  is the number of genuinely distinct columns it contains: columns that cannot be rebuilt as combinations of the others. *Full column rank*,  $\text{rank}(\mathbf{X}) = k$ , says no column is redundant: the data carry  $k$  separate pieces of information, one per coefficient.

## The OLS estimator

If  $\text{rank}(\mathbf{X}) = k$ ,  $\mathbf{X}'\mathbf{X}$  is invertible and

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbb{E}_n[X_i X_i'])^{-1} \mathbb{E}_n[X_i Y_i].$$

- Full rank needs  $n \geq k$ ; it is the matrix version of “ $X$  must vary.”
- Failures you will actually meet: the *dummy variable trap*; including a variable and a rescaling of it; a group indicator constant within a fixed effect.
- If rank fails, R silently drops a column and reports NA. That NA is never a mystery.

## Sanity Check: Recovering the Scalar Formula ★

Let  $k = 2$  with  $X_i = (1, X_i)'$ . Then

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Using  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$

$$\hat{\beta}_1 = \frac{n \sum_i X_i Y_i - \sum_i X_i \sum_i Y_i}{n \sum_i X_i^2 - (\sum_i X_i)^2} = \frac{\text{Cov}_n(X, Y)}{\text{Var}_n(X)}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Exactly what we derived by hand. The matrix form is a change of language, not of substance.

## Why Projection? It Is Tuesday's Argmin, Re-Read ★

Tuesday defined OLS by a minimization:  $\hat{\beta} = \arg \min_b \sum_{i=1}^n (Y_i - X_i' b)^2$ .

The one observation everything hangs on

A sum of squared entries is a squared *length*:  $\sum_i (Y_i - X_i' b)^2 = \|\mathbf{y} - \mathbf{X}b\|^2$ , the squared *distance* between two points of  $\mathbb{R}^n$ , the data  $\mathbf{y}$  and the candidate fit  $\mathbf{X}b$ . So the argmin says, word for word: **among all fits the model can produce, pick the one closest to the data.**

- The candidate fits  $\mathbf{X}b$  form a flat sheet, a “plane” (next frame), and “closest point on a plane” is a problem geometry solved long before calculus: **the foot of the perpendicular.**
- The right angle *is* the first-order condition: perpendicular to every column of  $\mathbf{X}$  means  $\mathbf{X}'\hat{\mathbf{e}} = \mathbf{0}$ , the normal equations, with no differentiation anywhere.
- Nothing new is computed: projection is the same argmin, *answered by geometry instead of calculus*. Next: the smallest possible case.

## What Does “All the Fits” Look Like? ★

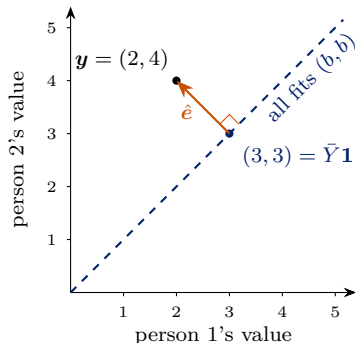
Read matrix-times-vector as a **recipe for mixing columns**:

$$\mathbf{X}b = b_1 \cdot (\text{column 1}) + b_2 \cdot (\text{column 2}) + \cdots + b_k \cdot (\text{column } k) :$$

each coefficient pours in that much of its column. The set of candidate fits is therefore *every possible mixture of the  $k$  columns*.

- **One column** (intercept only): all multiples of the single arrow **1**, a **line** through the origin. One knob to turn, one dimension of freedom.
- **Two columns**: all mixtures of two arrows, a flat **plane** through the origin. Two knobs, two dimensions.
- **$k$  columns**:  $k$  knobs, a  $k$ -dimensional flat sheet inside  $\mathbb{R}^n$ . That is *all* “ $k$ -dimensional subspace” means: **the set of points reachable by turning  $k$  independent knobs**.
- The data  **$y$**  generally sits *off* the sheet; the gap is the residual. And if one column is itself a mixture of the others, one knob is redundant and the sheet is thinner than  $k$ : **rank**, again.

# Where Projection Comes From: Two People First ★



- Two people, one variable:  $Y_1 = 2, Y_2 = 4$ . Plot the *dataset* as one point, one axis per person.
- The intercept-only model  $Y_i = b + e_i$  fits both people the same number: the point  $(b, b)$ . All candidates form the  $45^\circ$  line.
- Closest point on that line to  $(2, 4)$ : the perpendicular's foot,  $(3, 3)$ . So  $\hat{b} = 3 = \bar{Y}$ : **the mean is a projection**.
- Right angle, checked: residual  $(-1, 1)$ , line direction  $(1, 1)$ , dot product 0. Perpendicular = residuals sum to zero =  $(N1)$ .

With  $n$  people the picture lives in  $\mathbb{R}^n$ ; with regressors the line becomes the plane spanned by  $\mathbf{X}$ 's columns. OLS is always the same move: **drop the perpendicular**. (“Normal” equations: normal = perpendicular.)

## OLS Is a Projection ★

Think of  $\mathbf{y}$  as a single point in  $\mathbb{R}^n$ , one dimension per *observation*. The  $k$  columns of  $\mathbf{X}$  span a  $k$ -dimensional subspace

$$\text{col}(\mathbf{X}) = \{\mathbf{X}b : b \in \mathbb{R}^k\} \subset \mathbb{R}^n.$$

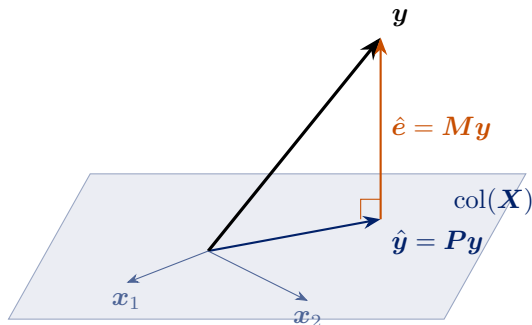
OLS finds the point in  $\text{col}(\mathbf{X})$  closest to  $\mathbf{y}$ :  $\hat{\mathbf{y}} = \arg \min_{v \in \text{col}(\mathbf{X})} \|\mathbf{y} - v\|^2$ .

### Projection and annihilator matrices

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{ ("hat matrix")}, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{P} \text{ ("residual maker")}$$

so that  $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$  and  $\hat{\mathbf{e}} = \mathbf{M}\mathbf{y}$ .

# The Picture ★



- $\hat{e} \perp \text{col}(\mathbf{X})$ , the geometric content of  $\mathbf{X}'\hat{e} = \mathbf{0}$ , which is the geometric content of “residuals are uncorrelated with the regressors.”
- Fitting is dropping a perpendicular. Everything else is bookkeeping.

# Properties of $P$ and $M$ ★

Symmetric

$$P' = P, \quad M' = M$$

Idempotent

$$PP = X(X'X)^{-1} \underbrace{X'X(X'X)^{-1}}_I X' = P$$

$$MM = (I - P)(I - P) = M$$

Mutually orthogonal

$$PM = P - PP = 0$$

For the traces:  $\text{tr}(P) = \text{tr}((X'X)^{-1}X'X) = \text{tr}(I_k) = k$ , using the cyclic property  $\text{tr}(ABC) = \text{tr}(BCA)$ , which we use repeatedly.

Act correctly on  $X$

$$PX = X, \quad MX = 0$$

Traces = dimensions

$$\text{tr}(P) = k$$

$$\text{tr}(M) = n - k$$

**Eigenvalues** are all 0 or 1;  $P$  has  $k$  ones,  $M$  has  $n - k$ .

# Leverage ★

The  $i$ th diagonal element of  $\mathbf{P}$ ,

$$h_{ii} = X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i = \frac{\partial \hat{Y}_i}{\partial Y_i},$$

is the **leverage** of observation  $i$ : how much unit  $i$ 's own outcome pulls its own fitted value.

- $0 \leq h_{ii} \leq 1$  and  $\sum_i h_{ii} = k$ , so average leverage is  $k/n$ .
- High leverage means unusual *covariate* values, not an unusual outcome.
- $\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii})$ : high-leverage points have artificially small residuals, they drag the line toward themselves.
- Leverage returns next week in the finite-sample corrections HC2 and HC3.

## Sums of Squares, Again ★

Because  $\mathbf{y} = \mathbf{P}\mathbf{y} + \mathbf{M}\mathbf{y}$  and  $\mathbf{P}\mathbf{M} = \mathbf{0}$ , Pythagoras gives

$$\mathbf{y}'\mathbf{y} = \mathbf{y}'\mathbf{P}\mathbf{y} + \mathbf{y}'\mathbf{M}\mathbf{y} = \hat{\mathbf{y}}'\hat{\mathbf{y}} + \hat{\mathbf{e}}'\hat{\mathbf{e}}.$$

Subtracting  $n\bar{Y}^2$  (valid when a constant is included) returns the scalar decomposition  $TSS = ESS + RSS$ .

- $R^2$  is also  $\cos^2$  of the angle between  $\mathbf{y} - \bar{Y}\mathbf{1}$  and  $\hat{\mathbf{y}} - \bar{Y}\mathbf{1}$ : fit is literally about angles.
- Adding a regressor enlarges  $\text{col}(\mathbf{X})$ , which can only move  $\hat{\mathbf{y}}$  closer to  $\mathbf{y}$ , the geometric reason  $R^2$  never falls.

## The Same Idea, Aimed at the Data: the SVD ★

OLS projected  $\mathbf{y}$  onto the closest point in  $\mathbf{X}$ 's column space. The **singular value decomposition** turns the same “closest fit” logic on the data matrix itself: any  $n \times k$  matrix factors as

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}',$$

where  $\mathbf{U}$  and  $\mathbf{V}$  hold perpendicular *directions* (through the observations and through the variables) and  $\mathbf{\Sigma}$  is diagonal with entries  $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ , the **singular values**.

- Read it as: every data matrix is a set of independent *directions of information*, ordered by how much of the data each carries.
- The rank from earlier = the number of nonzero singular values.
- **Keep only the top  $r$  directions**, and you have the rank- $r$  matrix *closest to  $\mathbf{X}$  in the least-squares sense*: the same “drop the perpendicular” logic as OLS, applied to the whole matrix rather than to  $\mathbf{y}$ .

# Latent Semantic Analysis: Projection Applied to Text ★

Let  $\mathbf{X}$  be a **document-term matrix**: one row per document, one column per word, entries counting occurrences. Thousands of columns; far fewer *themes*.

- Truncate the SVD at  $r \approx 100$  directions: each document becomes a mixture of  $r$  **latent semantic dimensions**, and words that co-occur across documents get pulled onto the same dimension. That is *latent semantic analysis* (Deerwester et al. 1990).
- Sociology runs on this geometry: Kozlowski, Taddy, and Evans (2019) *project* word vectors onto cultural axes such as rich-poor to track the meaning of class across a century, and the operation is literally the projection formula from a few frames back.
- Course connections: ridge regression shrinks along exactly these singular directions (Week 12), and synthetic control's factor model is a low-rank story about a panel (Week 11).

None of this is examinable: it is where the geometry goes when you point it at data sociologists care about.

## Unbiasedness, in Matrix Form ★

Tuesday's two-line scalar proof, now for any  $k$ :

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \epsilon) = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon,$$

$$E[\hat{\beta} \mid \mathbf{X}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \underbrace{E[\epsilon \mid \mathbf{X}]}_{=0} = \beta. \quad \blacksquare$$

- Note where the work is done: *only* by A4.
- Heteroskedasticity, clustering, and non-normality do **not** bias  $\hat{\beta}$ . They affect its *variance*, next week.
- A4 fails whenever an omitted variable, reverse causation, measurement error in  $X$ , or selection puts something correlated with  $X$  into  $\epsilon$ . That is the entire causal-inference problem in one sentence.

# The Gauss–Markov Theorem ★

## Theorem

Under A1–A5,  $\hat{\beta}$  is the **Best Linear Unbiased Estimator**: among all  $\tilde{\beta} = \mathbf{C}\mathbf{y}$  with  $\mathbf{C}$  a function of  $\mathbf{X}$  and  $E[\tilde{\beta} \mid \mathbf{X}] = \beta$ ,  $\text{Var}(\tilde{\beta} \mid \mathbf{X}) - \text{Var}(\hat{\beta} \mid \mathbf{X}) \succeq 0$ .

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**Proof.** Write  $C = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$ . Unbiasedness for all  $\beta$  forces  $\mathbf{D}\mathbf{X} = \mathbf{0}$ . Then

$$\begin{aligned}\text{Var}(C\mathbf{y} | \mathbf{X}) &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}] [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'] \\ &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} + \sigma^2 \underbrace{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}'}_{=0} + \sigma^2 \underbrace{\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{=0} + \sigma^2 \mathbf{D}\mathbf{D}' \\ &= \text{Var}(\hat{\beta} | \mathbf{X}) + \sigma^2 \mathbf{D}\mathbf{D}' \succeq \text{Var}(\hat{\beta} | \mathbf{X}). \quad \blacksquare\end{aligned}$$

# Partialling Out

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## The Frisch–Waugh–Lovell Theorem ★

Partition  $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$ : a treatment of interest and controls. Let  $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$  and define

$$\tilde{\mathbf{X}}_1 = \mathbf{M}_2\mathbf{X}_1, \quad \tilde{\mathbf{y}} = \mathbf{M}_2\mathbf{y}.$$

### Theorem

The OLS estimate  $\hat{\beta}_1$  from the full regression equals

$$\hat{\beta}_1 = (\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\mathbf{y},$$

and the residuals from the short regression are numerically identical to  $\hat{\mathbf{e}}$  from the full one.

This is the three-step recipe from earlier, now for any number of controls.

## Proof of FWL ★

Premultiply the model by  $M_2$  and use  $M_2X_2 = 0$ :

$$M_2y = M_2X_1\beta_1 + \underbrace{M_2X_2}_{=0}\beta_2 + M_2\epsilon \quad \implies \quad \tilde{y} = \tilde{X}_1\beta_1 + M_2\epsilon.$$

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For the estimator, decompose the fitted model  $y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{e}$  and premultiply by  $\tilde{X}'_1 = X'_1M_2$ :

$$\tilde{X}'_1y = \tilde{X}'_1\tilde{X}_1\hat{\beta}_1 + X'_1\underbrace{M_2X_2}_{=0}\hat{\beta}_2 + \underbrace{X'_1M_2}_{=0}\hat{e},$$

using  $M_2\hat{e} = \hat{e}$  (since  $X'_2\hat{e} = 0$ ) and  $X'_1\hat{e} = 0$ . Solving gives the result. Using  $\tilde{y}$  instead of  $y$  changes nothing because  $\tilde{X}'_1M_2 = \tilde{X}'_1$ . ■

# Why FWL Matters More Than It Looks

1. **Interpretation.** “Controlling for  $W$ ” = discarding all covariance between  $X_1$  and  $W$ . Your estimate lives on what remains.
2. **Computation.** Fixed-effects estimators *are* FWL: regressing on unit dummies is the same as demeaning within unit (Week 10).
3. **Diagnostics.** The added-variable plot is the only two-dimensional picture that faithfully represents a multivariate slope.
4. **Generality.** Nothing in the argument requires  $X_2$  to enter linearly, it only requires that we can compute the part of  $X_1$  that  $X_2$  predicts. That observation is the seed of a large literature, and we return to it briefly in Week 13.

## Omitted Variables, in Matrix Form ★

With  $\mathbf{X}_1$  included and  $\mathbf{X}_2$  omitted,

$$E[\tilde{\beta}_1 \mid \mathbf{X}] = \beta_1 + \underbrace{(\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' \mathbf{X}_2}_{\hat{\delta}} \beta_2.$$

- Each column of  $\hat{\delta}$  is the coefficient vector from regressing one omitted variable on all the included ones, exactly  $\delta_1$  from the scalar case, stacked.
- With several omitted variables the bias is a *sum* of terms which may offset, so “the bias must be positive” arguments require care.
- Sensitivity analysis (Week 6) formalizes the question: how strong would an omitted confounder have to be to overturn the result?

## Where This Leaves Us

- OLS minimizes squared errors. Everything else, the slope formula, the normal equations, FWL, OVB, follows from that one criterion by algebra.
- In the population, OLS estimates the best linear approximation to the CEF. Under A1–A4 the sample version is unbiased for it.
- Whether that coefficient is a *causal effect* is a separate question that regression cannot answer.
- **Next week:** we have  $\hat{\beta}$ ; now we need to know how uncertain it is, and the standard error your software prints is usually the wrong one.

### This week's reading

MHE Chapter 3; CCI Chapter 6. The lab practices the matrix algebra hands-on and computes OLS by hand in R.

# References

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